

ARUN ARUNACHALAM

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PROFESSIONAL SUMMARY

Applied AI/ML and GenAI executive with 17+ years architecting and delivering enterprise-scale artificial intelligence, machine learning, and data analytics platforms across capital markets at Bank of America. Deep practitioner expertise in GenAI (RAG pipelines, agentic AI), ML-based anomaly detection, and AI-assisted SDLC workflows, combined with a proven record of harnessing data and analytics to accelerate decision-making and drive measurable commercial outcomes. Adept at building and scaling Centers of Excellence-caliber capabilities — modernizing legacy infrastructure into cloud-native, event-driven, AI-enabled platforms that deliver data integrity, operational intelligence, and business impact at \$1T+ daily processing scale. Experienced collaborating with executive stakeholders to translate AI/ML strategy into production-grade solutions aligned with regulatory and compliance requirements.

SIGNATURE IMPACT

- **\$8M** enterprise transformation → **40% performance improvement, 30% cost reduction**
- **AI-assisted engineering** → **35% faster delivery, 45% productivity gain**
- Scaled platforms from 10 to **50+ systems** with **99.97% uptime**
- Enabled **~\$2B capital efficiency** via risk platform transformation

CORE CAPABILITIES

AI & GenAI Platforms (RAG Pipelines, Agentic AI) | Machine Learning & Anomaly Detection | Data Engineering & Analytics (Spark, Databricks, Snowflake, ETL/ELT) | AI-Assisted SDLC & Delivery Acceleration | Cloud-Native Architecture (AWS, Kubernetes) | Event-Driven Systems (Kafka) | Data Quality, Integrity & Governance | Trading & Risk Platforms | Platform Modernization & Transformation | Executive Stakeholder Leadership | Regulatory & Compliance Systems (FRB, FINRA, SEC)

PROFESSIONAL EXPERIENCE

Bank of America — Director, Global Risk & Capital Markets Technology (2015 - Present)

- Implemented AI-assisted development workflows and ML-based anomaly detection, delivering **45% productivity gain** and **35% faster delivery** across engineering teams
- Built and deployed GenAI platforms including RAG pipelines and agentic AI systems to accelerate intelligent decision-making and operational intelligence
- Led enterprise-scale platform transformation across trading and data systems supporting **\$1T+ daily processing**, ensuring data quality, integrity, and security at scale
- Modernized legacy systems into cloud-native, event-driven architecture (AWS, Kafka, Kubernetes), improving throughput by **40%** and achieving **99.97% uptime**
- Delivered real-time trading, pricing, and risk analytics platforms generating actionable insights for capital markets decision-making
- Drove AI-assisted SDLC workflows improving delivery speed, quality, and engineering velocity by **15%**

- Ensured regulatory compliance (FRB, FINRA, SEC) across mission-critical data and analytics systems
- Led global engineering teams (**30+ across US/UK/India**) executing large-scale data and AI platform programs

Bank of America — VP, Structured Products & Mortgage Technology (2011 - 2015)

- Built real-time pricing and analytics platforms for structured products and mortgage trading, enabling data-driven risk insights
- Led consolidation of risk systems contributing to **~\$2B capital efficiency** through improved data integrity and analytics accuracy
- Delivered ETL/ELT data pipelines supporting downstream analytics and reporting for trading desks

Bank of America — AVP, Whole Loan Trading Desk Support (prior to 2011)

- Architected and delivered end-to-end Java EE trading support applications with embedded analytics capabilities
- Introduced CI/CD practices reducing release cycle from monthly to weekly, accelerating delivery of data and analytics features

Senior Programmer Analyst — Knight Trading (2004)

DB2 / WebSphere Administrator & Sr. Java Developer — Kyocera Mita (2003)

Sr. Java Web Developer — Mellon Investor Services (2001)

Sr. Java GUI Developer — Thomson Financials (2001)

Sr. Java GUI Developer — Icor Brokerage (2000)

AI & INNOVATION

- Building GenAI platforms (RAG pipelines, agentic AI systems) for real-world enterprise applications, accelerating intelligent automation and decision support
- Applied ML for anomaly detection and operational intelligence across large-scale capital markets data systems
- Driving AI-assisted SDLC workflows improving delivery speed, quality, and engineering productivity
- Harnessing artificial intelligence and machine learning technologies to generate insights, enable data-driven decisions, and support commercial goals

EDUCATION

Bachelor of Technology — Anna University

SKILLS

AI & Data: GenAI (RAG pipelines, agentic AI), ML anomaly detection, AI-assisted SDLC, Spark, Databricks, Snowflake, ETL/ELT, Hadoop, Cassandra, MongoDB **Architecture:** Cloud-native architecture (AWS, Kubernetes), enterprise architecture, event-driven systems (Kafka), distributed systems, high-availability platforms (99.97% uptime) **Engineering:** Python, Java, Node.js, SQL, Oracle, Sybase, REST/SOA, Kafka, Kubernetes, AWS, WebLogic **Leadership:** Executive stakeholder management, global team leadership, P&L ownership, strategic planning, program management, SDLC, Agile, ITIL, SOX, FRB/FINRA/SEC compliance **Tools:** ServiceNow, JIRA, Git, Visio, Microsoft Project